

Pei (Patrick) Chen | 陈沛

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Professional Summary

Applied Scientist at Amazon working across the **full LLM mid/post-training stack** and the **data infrastructure** that feeds it, building **backbone models for agentic systems**. Hands-on experience with **100B+ parameter MoE** training on **Megatron** and **Verl/Slime**, spanning continual pre-training (ABF, YaRN) for **8K→128K context extension**, **SFT**, **DPO**, and **GRPO**. Built a **closed-loop data flywheel** for post-training: live-traffic failure mining, automated root-cause analysis, scalable annotation and quality control, and diversity- and difficulty-aware data selection. **Ph.D. in Computer Science** with **20+ publications** (8 first-authored) at **NeurIPS (Spotlight)**, **ACL**, **EMNLP**, and **NAACL**, one **US patent**, and area chair at **ARR**.

Experience

Amazon

2024.1 – Present

Applied Scientist (full-time)

Seattle, WA & Bay Area, CA

Led model training and system development for agentic LLM systems, with two focuses: (1) **context-aware LLM capabilities** for long-context modeling, retrieval grounding, multi-turn consistency, and personalization, and (2) **data-centric post-training** via closed-loop data flywheel systems for scalable and continuous improvement.

- **Amazon Rufus Foundation Modeling Team (startup-like) — Mid/Post-training for Agentic Backbone Models**: Co-led the context management and RAG workflow (4–5 junior Applied Scientists) for shopping agents, developing context-aware LLM capabilities on **100B+ MoE** models under real-world constraints.
 - Improved model faithfulness and grounding in retrieval-augmented settings through **synthetic IFT + DPO pipelines**, increasing factual accuracy from **67% to 90%** on customer-facing evaluations.
 - Advanced personalization via **verifiable reward design** and **GRPO**-based optimization, improving helpfulness from **84% to 93%** under user-specific preference settings.
 - Extended long-context capability from **8K to 128K** tokens through **continual pre-training (ABF, YaRN)**, enabling stable context window over extended multi-turn sessions.
 - Enhanced multi-turn context consistency by incorporating live-traffic failure patterns into **DPO** training, improving context carryover from **61% to 73%**.
- **Amazon Customer Service Team — Post-training Data Infrastructure & Flywheel**: Led the post-training data flywheel workflow (3–4 junior Applied Scientists); built and productionized a closed-loop **end-to-end post-training data infrastructure** for the team's multi-agent LLM system, spanning live-traffic failure mining, automated root-cause analysis, data generation and verification, and training/evaluation.
 - Built an **online failure mining and analytics system** from live customer service traffic, transforming raw failure signals into actionable patterns (e.g., identifying tone-related issues contributing to ~5% of weekly failures) that directly drove post-training data generation.
 - Designed and productionized a **scalable data annotation and quality control system** on Amazon Bedrock covering **SFT / DPO / GRPO / retrieval** data, with sample-level validation (rule-based + **LLM-as-judge**), dataset-level controls (**diversity, de-duplication**), and versioned management via **LakeFS** — adopted across teams for standardized training pipelines.
 - Developed **diversity-aware data selection**, improving agent tool-use accuracy from **86% to 93%**, while reducing training time by **4× (24h→6h)**.
 - Introduced **difficulty-aware curriculum data selection**, improving response generation quality from **88% to 91%**, while reducing training time by **5× (10d→2d)**.
- **Intern Mentorship**: Mentored two interns on LLM research projects in long-context modeling and multi-turn improvement with verifiable GRPO — **2 papers published, 1 under review**.

Amazon Web Services (AWS)

2023.5 – 2023.8

Applied Scientist (Intern)

Santa Clara, CA

- Researched LLM-based multi-agent systems for complex problem solving.

- Developed a **collaborative multi-agent** prompting framework that achieved state-of-the-art performance on complex science benchmarks in both zero-shot and few-shot settings.
- Published the [work](#) in Findings of NAACL 2024.

Amazon Web Services (AWS)

2022.6 – 2023.1

Applied Scientist (Intern)

Santa Clara, CA & remote

- Developed a novel **tabular language modeling** approach that represents tables as hypergraphs to better capture structural dependencies.
- Achieved state-of-the-art performance on tabular representation learning benchmarks.
- Published the [work](#) at NeurIPS 2023 as a Spotlight paper (**top 5%**).

Tencent AI Lab

2021.6 – 2021.8

NLP Researcher (Intern)

remote (College Station, TX)

- Built a comprehensive benchmark for zero-shot knowledge base completion, covering state-of-the-art methods and diverse knowledge sources.
- Resulted in a [paper](#) accepted to the ICDM 2022 Knowledge Graph Workshop.

Texas A&M University

2019.9 – 2023.12

Research Assistant and Teaching Assistant

College Station, TX

- Conducted research in NLP and information extraction, including named entity recognition, event extraction, and fine-grained opinion mining.
- Developed graph-based methods to model non-sequential dependencies among entity mentions, improving domain-specific NER and structured information extraction.
- Built datasets and benchmarks for fine-grained language understanding tasks, including location recognition.
- Served as a teaching assistant for CSCE 636 Deep Learning and CSCE 313 Computer Systems.

National Lab of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences

2018.1 – 2019.7

Research Engineer (Intern & Full-time)

Beijing, China

- Built information extraction and evaluation systems for financial event extraction and causality detection, bridging NLP research and real-world financial text applications.

Innovation Lab of Global Exchange, State Street

2017.7 – 2018.1

Data Analyst (Intern)

Hangzhou, China

- Supported data cleaning, analysis, visualization, and database construction for financial applications.

Selected Publications

LLM & Foundation Model Training

Research on **data-centric and structure-aware LLM training**, spanning **architectural innovations** (e.g., set-attention for tabular LMs), synthetic instruction tuning, context denoising, inductive reasoning, and scalable preference alignment.

- **Pei Chen**, Soumajyoti Sarkar, Leonard Lausen, et al. "[HYTREL: Hypergraph-enhanced Tabular Data Representation Learning](#)." **NeurIPS 2023, Spotlight (5%), first work to use set-attention for tabular LMs.**
- Ming Li, **Pei Chen**, Chenguang Wang, et al. "[Mosaic-IT: Cost-Free Compositional Data Synthesis for Instruction Tuning](#)." ACL 2025, Findings.
- Hongye Jin, **Pei Chen**, Jingfeng Yang, et al. "[END: Early Noise Dropping for Efficient and Effective Context Denoising](#)." pre-print.
- Wangtao Sun, Haotian Xu, Xuanqing Yu, **Pei Chen**, et al. "[ItD: Large Language Models Can Teach Themselves Induction through Deduction](#)." ACL-2024, long paper.
- Zhaoxuan Tan, Zheng Li, Tianyi Liu, Haodong Wang, **Pei Chen**, et al. "[Aligning Large Language Models with Implicit Preferences from User-Generated Content](#)." ACL 2025, long paper.

Agent

Research on **agentic LLM capabilities**, combining multi-agent reasoning, agent-oriented pre-training, guided function calling, and **verifiable** trajectory-based tool-use training for scalable and reliable agent systems.

- **Pei Chen**, Shuai Zhang, Boran Han. [“CoMM: Collaborative Multi-Agent, Multi-Reasoning-Path Prompting for Complex Problem Solving.”](#) NAACL 2024, long paper, **early work in demonstrating multi-agent systems for complex LLM reasoning.**
- Yuchen Zhuang, Jingfeng Yang, Haoming Jiang, Xin Liu, **Pei Chen**, et al. [“Hephaestus: Improving Fundamental Agent Capabilities of Large Language Models through Continual Pre-Training.”](#) NAACL 2025, long paper.
- Hy Dang, Tianyi Liu, Zhuofeng Wu, Jingfeng Yang, **Pei Chen**, et al. [“Improving Large Language Models Function Calling and Interpretability via Guided-Structured Templates.”](#) EMNLP 2025, long paper.
- Ziyi Wang, Yuxuan Lu, Yimeng Zhang, **Pei Chen**, et al. [“Trajectory2Task: Training Robust Tool-Calling Agents with Synthesized Yet Verifiable Data for Complex User Intent.”](#) accepted to ACL 2026, long paper.

RAG & Multi-turn & Personalization

Research on **context-aware conversational LLMs**, spanning long-context modeling, conversational retrieval, multi-turn consistency, and scalable personalization.

- **Pei Chen**, Hongye Jin, Cheng-Che Lee, et al. [“LongLeader: A Comprehensive Leaderboard for Large Language Models in Long-context Scenarios.”](#) NAACL 2025, long paper.
- Ming Li, **Pei Chen**, Zhenhao Zhang, et al. [“Mitigating Lost in Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards.”](#) Accepted to ACL 2026, long paper.
- Yichuan Li, Xinyang Zhang, Chenwei Zhang, Mao Li, **Pei Chen**, et al. [“ALERT: An LLM-powered Benchmark for Automatic Evaluation of Recommendation Explanations.”](#) NAACL 2025, long paper.
- Fengran Mo, Yifan Gao, Chuan Meng, Xin Liu, **Pei Chen**, et al. [“UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations.”](#) ACL 2025, long paper.
- Fengran Mo, Yifan Gao, Zhuofeng Wu, Xin Liu, **Pei Chen**, et al. [“Leveraging historical information to boost retrieval-augmented generation in conversations.”](#) Information Processing & Management, March 2026.
- Zhaoxuan Tan, Zixuan Zhang, Haoyang Wen, Zheng Li, **Pei Chen**, et al. [“Instant Personalized Large Language Model Adaptation via Hypernetwork.”](#), accepted to ACL 2026.
- Yizhuo Chen, Xin Liu, Ruijie Wang, Zheng Li, **Pei Chen**, et al. [“POPI: Personalizing LLMs via Optimized Natural Language Preference Inference.”](#) pre-print.

Information Extraction

Research on **structured information extraction and graph-based modeling**, capturing non-sequential dependencies for cross-sentence entity and event understanding in noisy, domain-specific text, forming foundations for robust knowledge grounding and structured memory in LLM systems.

- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. [“Explicitly Capturing Relations between Entity Mentions via Graph Neural Networks for Domain-specific Named Entity Recognition.”](#) ACL-IJCNLP 2021, short paper.
- **Pei Chen**, Haotian Xu, Cheng Zhang, and Ruihong Huang. [“Crossroads, Buildings and Neighborhoods: a Dataset for Fine-grained Location Recognition.”](#) NAACL 2022, long paper.
- **Pei Chen**, Hang Yang, Kang Liu, et al. [“Reconstructing Event Regions for Event Extraction via Graph Attention Networks.”](#) AACL-IJCNLP 2020, long paper.
- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. [“Domain-specific Named Entity Recognition via Graph Neural Networks.”](#) US Patent 12,530,529, 2026.

Education

Texas A&M University
Ph.D. in Computer Science

2019.8 - 2024.5
College Station, TX

- **GPA:** 4.0/4.0
- **Research Directions:** NLP and information extraction (NER, event extraction, knowledge base completion); later extended to LLMs and agentic systems.

- **Dissertation:** Connecting the Dots: Improving Information Extraction by Modeling the Non-Sequential Dependencies of Entity Mentions.

Southwestern University of Finance and Economics

Master's Degree in Finance

2016.9 – 2018.6

Chengdu, China

- **GPA:** 3.9/5.0, ranking 1/178.
- **Thesis:** Does News Sentiment Predict the Stock Market? An Example on Chinese Growth Market.
- **Awards:** 2017 National Scholarship for Graduate Student.

National University of Defense Technology

B.Eng. in Simulation Engineering

2010.9 – 2014.6

Changsha, China

- **GPA:** 88.61/100, ranking 1/45.
- **Thesis:** Analyze and reconstruct the multi-resolution modeling technology of a simulation system written by millions of lines of C++ code.

Professional Service

- 2026: **Area Chair for ARR**, Program Committee/Reviewer for ARR, ICLR 2026, ICML 2026.
- 2025: Program Committee/Reviewer for ARR, COLING 2025, ICLR-2025, NeurIPS 2025
- 2024: Program Committee/Reviewer for ARR, NeurIPS 2024, TPAMI
- 2023: Program Committee/Reviewer for ACL, EMNLP, ARR
- 2022: Program Committee/Reviewer for EMNLP, ARR, NLPCC
- 2021: Program Committee/Reviewer for EMNLP, ARR, NLPCC

Skills

- **Training Frameworks:** Megatron, Verl, Slime, Weights & Biases
- **Post-training Methods:** Continual Pre-training/Mid-training (ABF, YaRN), SFT, DPO, GRPO, DAPO
- **Data Infra:** LakeFS, Synthetic data generation, LLM-as-judge validation
- **Technical Focus:** Long-context modeling, preference alignment, multi-agent system design, RAG, hallucination mitigation, data flywheel
- **Programming:** Python, PyTorch, SQL, C/C++, Git, Docker